

Principles of Computer Security, Sixth Edition

Chapter 10 Solutions

Key Terms Quiz

1. router
2. network access control (NAC)
3. network interface card (NIC)
4. bridge, switch
5. Sandboxing
6. Workstation
7. hub
8. shielded twisted-pair (STP) or unshielded twisted-pair (UTP)
9. firewall
10. private branch exchange (PBX)

Multiple-Choice Quiz

1. **C.** Switches operate at Layer 2, the data link layer of the OSI model.
2. **D.** The standard connector for UTP in an Ethernet network is the RJ-45 connector. An RJ-45 connector is larger than a standard phone connector.
3. **C.** A coaxial connector carries one wire, one physical circuit.
4. **A.** Microsoft NAP (Network Access Protection) is one form of network access control.
5. **C.** The twist in twisted-pair wires reduces crosstalk between wires.
6. **B.** VM sprawl is a condition of uncontrolled spreading of disorganization caused by a lack of an organizational structure when many similar elements require management. Just as you can lose a file or an e-mail and have to go hunt for it, you can misplace a virtual machine.
7. **D.** The Simple Network Management Protocol is used to control network devices from a central control location.

8. **D.** Firewalls can do all of these things.
9. **A.** SMTP, the Simple Mail Transfer Protocol, is used to move e-mail across a network.
10. **D.** USB-based flash memory is low cost, fast, and high capacity—currently up to 1 TB.

Essay Quiz

1. Routers allow different IP address ranges to talk to each other. Routers are used for global addressing and operate off of the IP address (end-to-end addressing). They are more expensive than switches. Switches allow communication between machines on the same IP subnet. They work on Layer 2 addresses and are next hop only in field of view. Switches are for terminal local addressing inside local networks.
2. The following are common threats to the transmission media in a network, organized by type of transmission media:

Media Type	Threats
Cabled media (coax, UTP, STP)	Cable interruption (such as being cut by a backhoe) Loose connections Tapping for eavesdropping
Wireless	Eavesdropping RF interference
IR	Eavesdropping Short-range issues, such as blocked line of sight
Fiber-optic cable	Cable interruption (such as being cut by a backhoe)